

Mohit Marvania

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RESEARCH INTERESTS

LLM alignment robustness – specifically whether safety and capability properties survive adversarial stress-testing, sequential fine-tuning, and distribution shift. Interests span reinforcement learning from human feedback, empirical AI safety evaluations, and scalable oversight.

EDUCATION

George Mason University <i>M.S. in Computer Science – Machine Learning</i>	Fairfax, Virginia <i>Expected, May 2026</i>
Charotar University of Science & Technology <i>Bachelor of Technology in Computer Science (GPA: 3.78/4.0)</i>	Changa, India <i>Graduated, April 2024</i>

RESEARCH EXPERIENCE & PUBLICATIONS

Graduate Research Assistant <i>Supervisor: Dr. Abolfazl Safikhani & Dr. Ziwei Zhu, George Mason University.</i>	Fairfax, VA <i>Jan 2025 – Present</i>
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Catastrophic Forgetting in Continual Preference Alignment of LLMs

<i>Co-First Author Advisor: Dr. Ziwei Zhu</i>	<i>Targeting (ARR May 2026)</i>
- Found that safety alignment degrades silently during sequential fine-tuning, and that the degradation pattern depends on task ordering and algorithm choice more than model size. First systematic comparison of PPO, DPO, and GRPO across 5 alignment dimensions and 11 task orderings. - Owned the complete GRPO implementation end-to-end, reward model training, policy fine-tuning on Llama-3.1-8B and Qwen3-4B, BWT metrics evaluation across all sequence-model combinations on Slurm-managed GPU clusters.	

Stability-Aware Feature Design for Robust Watermark Detection in Machine-Generated Text

<i>Co-First Author Advisor: Dr. Abolfazl Safikhani</i>	<i>Accepted (ICML 2026)</i>
- Identified a critical failure mode in LLM watermarking: standard detectors collapse below 60% AUC under recursive paraphrasing, a real-world attack any adversary can run. Built PSS framework that holds at 91.2% AUC at depth-8 and greater than 83.7% AUC across unseen LLMs, paraphrasers, and domains without retraining. - Led paper writing and co-designed the adversarial evaluation pipeline using Llama-3 and Qwen2 as watermarked LLMs with Mistral-7B, Qwen2, and Gemma-7B paraphrasers, optimized detection to O(n) complexity via rolling window analysis on A100s.	

VeriSim: A Configurable Framework for Evaluating Medical AI Under Realistic Patient Noise | [\(Paper link\)](#)

<i>Co-First Author Advisor: Dr. Yikuan Li & Dr. Ziwei Zhu</i>	<i>Targeting (ARR May 2026)</i>
- Showed that SOTA medical LLMs lose 15-25% diagnostic accuracy when patient simulators behave like real patients, not idealized benchmark inputs. Current static benchmarks are measuring something different from real-world performance. - Designed the patient simulator, 6-pillar clinical noise injection framework, and LLM-as-a-Judge along with human evaluation pipeline (Cohen’s Kappa=0.81 human agreement).	

Reasoning Under Constraint: How Batch Prompting Suppresses Overthinking | [\(Paper link\)](#)

<i>Contributor Independent Collaboration</i>	<i>Accepted (ICLR 2026 Workshop on LLM Reasoning)</i>
Batch prompting cuts reasoning tokens 76% while preserving or improving accuracy across 13 benchmarks, an inference-time technique that requires no model modification, applicable to closed APIs like OpenAI-o1.	

Machine Learning Researcher Intern

<i>Supervisor: Dr. Amit Thakkar, Charusat University.</i>	Changa, India <i>Dec 2023 – Apr 2024</i>
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YOLOv8-based Emotion Recognition for Student Engagement Assessment

<i>First Author</i>	<i>Accepted (AIP Conference Proceedings, 2025)</i>
Built and published real-time engagement detection system achieving 72.3% accuracy across 6 behavioral classes, automated dataset fusion pipeline (4000+ samples) cutting preprocessing time by 80%.	

INDUSTRY EXPERIENCE

Software Engineer Intern

<i>Raven Technolabs</i>	Rajkot, India <i>May 2023 – Jul 2023</i>
Reduced database write latency 30% for AI-assisted receipt system serving 2k+ weekly users via batched operations, introduced CI/CD pipelines that cut post-release hotfixes.	

SKILLS

Languages: Python (Advanced), C++, Java, JavaScript/TypeScript, Bash, SQL
RLHF & Alignment: PPO, DPO, GRPO, reward modeling, preference optimization, continual alignment, scalable oversight
ML & Safety Stack: PyTorch, HuggingFace, Transformers, TRL, DeepSpeed, Unsloth, Weights & Biases
Models: Llama-3, Mistral, Qwen, DeepSeek-R1, GPT-4o API, Claude API
Infrastructure: Slurm (HPC cluster management), Docker, AWS (EC2/S3), Linux (Ubuntu)